

	De-Havilland DHC-8-400 B1 & B2 TRAINING MANUAL		COURSE CODE		DGM12CBXX3Q4PW	
			ISSUE	02	DATE	20 Sep 2024
			REVISION		DATE	



CHAPTER

ATA-100

PUBLICATIONS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY.

CONTENTS

CONTENTS	2
COURSE OBJECTIVES	3
AIRCRAFT MAINTENANCE MANUALS.....	4

COURSE OBJECTIVES

Introduction

1. Upon completion of the course, you will be able to identify safety precautions related to the airframe, systems and powerplant.
2. Identify maintenance practices related to the airframe, systems and powerplant.
3. Define the general layout of the aircraft's major systems including terminology, and nomenclature.
4. Identify special tooling and test equipment.
5. Demonstrate knowledge of the main ramp and transit activities including servicing.
6. Describe in detail systems and aircraft handling.
7. Explain the normal and abnormal functioning of each major system, including fault diagnostics.
8. Demonstrate proficiency in the use of documentation (MEL/CDL) to determine aircraft airworthiness.
9. Identify and use appropriate documentation.

AIRCRAFT MAINTENANCE MANUALS

General

The Aircraft Maintenance Manual (AMM), for the Dash 8 Q400, is in agreement with the Air Transport Association of America Specification No. 100 (ATA-100), revision 34, dated February 15, 1996. The AMM contains two parts:

- Part 1 - System Description Section
- Part 2 - Practices and Procedures.

The information given in this introduction is applicable to Part 2 of the AMM, Practices and Procedures, reference is made to the SDS as necessary.

The AMM Part 2 gives instructions to service, replace, adjust, test, inspect, check, clean, paint, and repair the aircraft systems and equipment. The AMM gives structure inspection criteria, however, instructions to repair the structure are in the DHC8 402 (Q400) Structural Repair Manual (PSM 1-84-3). Instructions and data to inspect and repair equipment off the aircraft are contained in vendor Component Maintenance Manuals or the DHC8 402 (Q400) Component Maintenance Manual (PSM 1-84-6).

If the data or procedures written in this manual are different from those given by the regulatory authority, obey the data and procedures of the regulatory authority.

Organization of Manual

The maintenance manual is organized in levels as specified by ATA-100. These levels are usually referred to as a chapter, section, and subject. Each page contains the chapter-section-subject number and revision date of the page.

Chapter/System

This level is usually known as the chapter. It has inter-related components arranged to do a specific function: It is further divided into subsystems usually known as sections. Each chapter is assigned its number and title by ATA-100 and is identified by the first element in the standard numbering system. The table that follows shows the chapter titles and their related numbers.

Section/Subsystem

This level is usually known as the section. It permits the chapter to be divided into subsystems and is identified by the second element in the standard numbering system.

Subject/Unit

This level is usually known as the subject. It permits the identification of individual units in a system or section and is identified by the third element in the standard numbering system.

Standard Numbering System

The manual numbering system is specified by ATA-100. It is a three-element numbering system that divides the manual into chapter/section/subject. Each element has two digits and is given as follows:

Element

Definition of Element

1 2 3

34-00-00 Material applicable to the system as a whole

└ System 34-00-00 Navigation

34-10-00 Material applicable to the subsystem as a whole

└ Subsystem 34-10-00 Flight Environmental Data

34-11-00 Material applicable to the sub-subsystem as a whole

└ System 34-11-00 Air Data System

34-11-11 Material applicable to a specific unit of the sub-subsystem

└└ System 34-11-11 Probes, Static Temperature

Page Block Assignment

The subjects in the maintenance manual are divided into page blocks with the headings specified by ATA-100. The page blocks are identified as follows:

Heading	Page Numbers
Systems Description Section	1 to 99
Component Location	101 to 199
Maintenance Practices	201 to 299
Servicing	301 to 399
Removal/Installation	401 to 499
Adjustment/Test	501 to 599
Inspection/Check	601 to 699
Cleaning/Painting	701 to 799
Repairs	801 to 899

If additional pages are necessary, the page numbers stop at 98 and continue with 98.1, 98.2, 98.3, etc.

The page blocks are usually written independently, but if page blocks are short, they are combined into one heading: Maintenance Practices, page block 201 to 299. General maintenance instructions, that are not identified by one of the heading given above, are put into the Maintenance Practices, page block 201 to 299.

System Description Section (1 to 99)

The System Description Section comprises Part 1 of the Aircraft Maintenance Manual. This section gives the function, operation, configuration, and control of the system and components in the aircraft. The information is given in sufficient detail to let the technician know the function and construction of the system.

Maintenance Practices (201 to 299)

This page block is used when:

- The content of individual pages is brief and it can be combined into one topic
- The procedure does not agree with any specific block assignment.

Servicing (301 to 399)

Servicing page blocks are used in Chapter 12 and system chapters.

1. Chapter 12 contains scheduled and unscheduled servicing applicable to the whole aircraft. Included in this chapter are replenishment of items such as fuel, oil, hydraulic fluid, water, tire pressure, etc.
2. Servicing which is necessary to complete other maintenance tasks is found in related chapters. This includes tasks such as bleeding a component.

Removal/Installation (401 to 499)

- (a) Removal/installation procedures have two independent tasks, one for the removal and one for the installation. However, the job set-up and close out procedures are not given in both tasks.
- (b) The job close out procedures are not given in the removal task because the installation usually follows it. If the installation task does not immediately follow the removal task, it is assumed that the operator will use the applicable close out procedures.
- (c) The step "Make sure that the aircraft is in the same configuration as in the removal task", is given as a reminder to the operator that the aircraft configuration, from steps done in the removal task, is still the same. These steps can include such items as:

- Removal of hydraulic power
- Circuit breaker adjustment
- Removal of access panels
- Safety practices/precautions

Adjustment/Test (501 to 599)

- (a) There are three types of tests: operational, functional, and system. The tests are specified in ATA-100 as follows:

Operational Test

That procedure required to ascertain only that a system or unit is operable. These tests should require no special equipment or facilities other than those installed on the aircraft and should be comparable to the tests done by the flight crews. It is not intended that the operational test of the unit shall meet the specifications and tolerances ordinarily established for overhaul or major maintenance periods.

Functional Test

That procedure required to ascertain that a system or unit is functioning in all aspects in accordance with minimum acceptable system or unit design specifications. These tests may require supplemental ground support equipment and should be more specific and detailed than an operational test. It should contain all necessary information to perform proficiency tests to maintain system or unit reliability at an acceptable level, without reference to additional documents.

System Test

That procedure containing all adjustment specifications and tolerances required to maintain system and/or unit performance at maximum efficiency and design specifications. It shall be self-contained and may duplicate other tests. It is normally used at major maintenance periods.

Inspection/Check (601 to 699)

(a) This page block contains information and instructions to inspect or perform a check of:

- System
- Component
- Area (zone)
- Connected parts

(b) Inspection/check specifies the rejection criteria for the related equipment. The categories of inspection/check are as follows:

- Inspection/Check on the Aircraft
 - Visual inspections or checks for cracks, damage, corrosion, etc.
 - Inspection or checks with measurement of displacement, travel, clearances, rate of leakage, etc.
- Inspection/Check of Component Removed from Aircraft
 - This inspection/check category includes visual inspections of components that are removed from the aircraft and the connections between the aircraft and the removed components.

Cleaning/Painting (701 to 799)

This page block gives the necessary procedures to clean and/or paint a part or area of the aircraft.

Repairs (801 to 899)

This page block gives all approved repair procedures for the aircraft. It does not include those repairs contained in the Component Maintenance Manuals and the Structural Repair Manual.

Tasks

All page blocks in Part II of the AMM contain tasks. A task is a complete procedure. It can also contain references to other tasks where you can find related procedures. A task will contain a general paragraph, one or more topics, and one or more subtasks. Tasks usually have an illustration. The tasks in a page block are identified with a task number and title.

Types of Tasks

Tasks are identified with a task number that contains the chapter number, the section number, the subject number, the AMTOSS function code number, and a sequence number. A full description of the Aircraft Maintenance Task Oriented Support System (AMTOSS) numbering system is given later in this Introduction.

General Paragraph

The general paragraph in a task gives the purpose of the task, the location of the component and other general information about the task. The general paragraph will not contain procedural steps or cautions and warnings.

Maintenance Topics

Topics are high level headings used to divide the task into groups of subtasks (steps). They permit recognition of major parts of the task. One or more topics are always contained in a task and are given in the sequence that follows:

- Job Set-Up Information
- Job Set-Up
- Procedure
- Close Out

These are the usual topics found in an AMM task. Other topics are used as applicable.

Effectivity

The procedures that apply only to specified optional equipment will have a statement at the start of the task to identify the related aircraft. This statement is known as an effectivity statement.

In an effectivity statement, an aircraft tail number is given in a four digit number and begins with the number “4” (Example: 4XXX). The “4” represents the Q400 aircraft series and XXX represents the sequentially assigned production serial number to the aircraft. An effectivity statement will be at the start of all tasks. The statement will list the applicable aircraft or give the statement “ALL” if the task applies to all aircraft.

An EFFECTIVITY box at the bottom of the left corner of each page gives the page where the task effectivity started.

Task Procedures

The steps of the procedure are written in a logical order as interpreted by the technical writer and based on the fact that the aircraft is in the standard configuration for maintenance.

The standard configuration for maintenance is as follows:

- The aircraft is parked
- The engines are stopped
- The wing flaps are fully retracted
- There is no electrical or hydraulic power applied to the aircraft
- All circuit breakers are closed.

If a different configuration is necessary to do the procedure, these steps will be included in the Job Set-Up topic. For example:

- Energize the aircraft electrical system
- Open and tag the circuit breakers that follow:
- Remove the access panels that follow:
- Make sure that the aircraft is in the same configuration as in the removal task.

It is not necessary for a task to refer to the Standard Configuration for Maintenance task. All tasks are written with the assumption that the aircraft is in the standard configuration for maintenance before you start the task.

Standard Practices

Tasks that contain standard practice procedures are usually found in Chapters 20 and 51. Chapter 20 contains standard practice tasks related to aircraft systems (Example: Installation of Preload Indicating Washers, Repair of Hydraulic Lines with Permaswage Fittings). Chapter 51 contains standard practice tasks related to the aircraft structure (Example: Application of Sealant on External Paint Edges, Application of F19 Epoxy Polyamide Primer).

Torque values to tighten fasteners are only given in the text of procedures when a special torque value is necessary. When a procedure gives instructions to install a fastener and no torque value is given, use the standard practices

System Diagnostic Tasks

Some tasks give procedures to get access to the data and codes from built-in tests of the electronic control units in the aircraft systems. Systems use one of three different methods to show this data:

- Codes or data are shown on the avionics radio control display unit (ARCDU) through the central diagnostic system (CDS) (Example: Timer Monitor Control Unit (TMCU))
- Codes or data are shown on the engine display (ED) through the central diagnostic system (CDS) (Example: FADEC)
- Codes or data are shown on the display panel on the electronic control unit (example: Proximity Switch Electronic Unit (PSEU))

Definition of Warning, Caution, and Note

ATA-100 defines warnings, cautions, and notes as follows:

WARNING

Calls attention to the use of materials, processes, methods, procedures, or limits which must be followed precisely to prevent injury to persons.

CAUTION

Calls attention to methods and procedures which must be followed to prevent damage to equipment.

NOTE

Calls attention to methods that make the job easier.

Consumable Materials

Some of the maintenance procedures in the AMM have consumable materials tables. These tables identify the consumable materials that are necessary to do the maintenance. A consumable material is an item that is used once and is subsequently replaced

The Consumable Products Manual, PSM 1–84–18, supplies lists which identify all of the consumable materials that are used in the AMM. The consumable materials that are given in the Consumable Products Manual are divided into the sections given in the table that follows:

REFERENCE	DESIGNATION
01–XX	Fuels
02–XX	Hydraulic Fluids
03–XX	Oils
04–XX	Greases and Lubricants
05–XX	Adhesives
06–XX	Sealants
07–XX	Protective Coatings and Primers
08–XX	Solvents and Cleaning Agents
09–XX	Strippers
10–XX	Paints and Lacquers
11–XX	Anti-Icing and Deicing Materials
12–XX	Disinfectants and Additives
10–XX	Paints and Lacquers
11–XX	Anti-Icing and Deicing Materials

The first two digits of the REFERENCE number identify the section where the maintenance technician can find the consumable material. The last two digits are the sequence number for the specific consumable material.

AMTOSS (AIRCRAFT MAINTENANCE TASK ORIENTED SUPPORT SYSTEM) TASK Numbering System

A. General

The function of AMTOSS is to improve the organization of the AMM and to permit the use of an automated data retrieval system. It uses a standard and unique numbering system to give a numerical identity to maintenance Tasks usually done in an airline maintenance program.

B. Numbering System

(1) The numbering system is an expansion of the ATA-100 three-element numbering system and has seven elements. Maintenance procedures are divided into Tasks.

(2) The numbers are defined as follows:

ELEMENT NUMBER	TASK
	29-00-01-400-801-A00-XXX
	1 2 3 4 5 6 7
1, 2, & 3	Chapter-Section-Subject
4	This three digit numerical function code identifies/defines the maintenance being performed. The definition of all function codes can be found in the AMTOSS Function Code Index.
5	This three digit unique numerical identifier lets similarly identified TASKS be unique when elements 1 to 4 are the same.
6	This element is not used unless necessary. Element 6 is used to identify a TASK effectivity or to identify sheet numbers for illustration Task numbers. When element 6 is used to identify a TASK effectivity, the code is alphanumeric and starts with A00, B00, etc. The first digit is alphabetic. It identifies the differences in procedure within the TASK due to the incorporation of modifications, service bulletins, etc. When element 6 is used to identify a sheet number, the code is numeric and starts at 001, 002, etc.
7	This element is not used unless necessary. This three digit alphabetic identifier is used to identify airline unique data. The airline is responsible for the identified TASK content.

AMTOSS Function Code Index

AMTOSS CODE	FUNCTION
000	Removal
100	Cleaning
200	Inspection, Check
300	Correction, Repair
400	Installation
500	Material and Aircraft Handling
600	Servicing/Preserving/Lubricating
700	Testing, Checking
800	Miscellaneous
900	Change = Remove + Install

Related Manuals

Name of Manual	Abbreviation
Aircraft Illustrated Parts Catalog	AIPC
Component Maintenance Manual	CMM
Consumable Products Manual	CPM
Illustrated Tool and Equipment Manual	ITEM
Fault Isolation Manual	FIM
Maintenance Requirements Manual	MRM
Maintenance Planning Document	MPD
Maintenance Task Card Manual	MTCM
Master Minimum Equipment List	MMEL
Minimum Equipment List Procedures	MELP
Non-Destructive Testing Manual	NDTM
Power Plant Build-Up Manual	PPBM
Structural Repair Manual	SRM
Weight and Balance Manual	WBM
Wiring Manual	WM

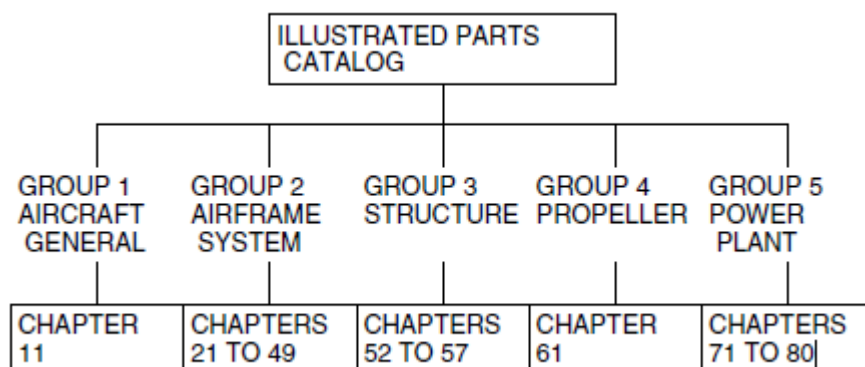
Aircraft Illustrated Parts Catalog (AIPC)

The AIPC lists, describes and illustrates assemblies, subassemblies and detailed parts of the Dash 8. It is intended for use in requisitioning, issuing, identifying parts, and in illustrating assembly relationships.

The AIPC does not contain data related to the overhaul of the components or the repair of the aircraft structure. Data to overhaul a component can be found in the Component Maintenance Manual

The AIPC is a companion document to the Aircraft Maintenance Manual (AMM) and includes all parts for which a maintenance practice has been provided.

The AIPC is organized on five levels as specified by ATA 100. They are usually referred to as group, chapter, section and subject.



Fault Isolation Manual (FIM)

To avoid or reduce dispatch delays, the FIM gives the original equipment manufacture's (OEM) recommended procedures to quickly and effectively isolate discrepancies in the Q400 aircraft.

Maintenance personnel in different operating environments can easily use the FIM. It uses electronic fault isolation built in test equipment (BITE). It gives the technical data the maintenance personnel need to isolate the cause of the discrepancy and to give the corrective action. The FIM is compatible with airline concepts, maintenance training, and the tasks in the aircraft maintenance manual.

The FIM is organized in the three sections that follow:

- Introduction
- FIM Tasks
- Message/Fault Lists

The FIM does not give procedures for recurring defects which are defects.

System Schematic Manual (SSM)

The system schematics show the interconnection of mechanical, electrical and electronic equipment and components of systems, subsystems and sub-subsystems and/or functions. The manual presents the systems in sufficient detail to permit fault isolation and understanding of system operations by maintenance personnel. Clarity of presentation of the information provided on the system schematics is the prime requirement. The SSM is presented to show the mechanical and/or electrical or electrical/mechanical schematics in a simplified manner for ease of understanding.

The SSM provides simplified schematics of all sub-subsystem and is presented in such a way as to explain the overall operation of a system with interface to other systems. As required, these schematics contain electrical and/or mechanical, inclusive of abstract renderings and be consistent to orient the user to the system in question.

Minimum Equipment List Procedure (MELP)

This Minimum Equipment List Procedures Manual is a companion document to the Department of Transport (Canada) Master Minimum Equipment List, and is for use by both flight crew and maintenance personnel.

It reiterates the conditions in the MEL which must be met when any MEL item is unserviceable.

The following information is contained in MELP:

- Identifying (placarding) unserviceable MEL items.
- Additional operating procedures, as required, for each (unserviceable) MEL item.
- Any required deactivation and other maintenance procedures. Tools required for these procedures are listed as applicable.

Where an "(M)" requirement has been identified for a particular MEL item, a specific maintenance procedure must be accomplished prior to operation with the listed item inoperative. Normally these procedures are accomplished by maintenance personnel; however, other personnel may be qualified and authorized to perform certain functions.

Where an "(M#)" requirement has been identified for a particular MEL item, a specific maintenance procedure must be accomplished prior to operation with the item inoperative. This procedure is accomplished by maintenance personnel.

Where an "(O)" requirement has been identified for a particular MEL item, a specific operations procedure must be accomplished in planning for and/or operating with the listed item inoperative. Normally these procedures are accomplished by flight crew; however, other personnel may be qualified and authorized to perform certain function.

Corrosion Prevention and Control

Corrosion is a serious and costly problem. The main concern is for the aircraft's structural integrity throughout its projected life. Environment and poor maintenance practices combine to make materials particularly vulnerable to the insidious nature of corrosion. Unchecked corrosion can cause structural and system failures and, in extreme cases, loss of aircraft. It is necessary to recognize corrosion in its early stages and restore it to a serviceable condition by simple repair procedures. A corrosion prevention and control program is very important to check corrosion before it becomes an airworthiness concern.

Airworthiness Limitation Items (ALI)

To meet Type certification requirements the certification program identifies maintenance tasks where the task must be performed at or before the required interval.

Five types of maintenance tasks are included in the ALI:

Certification Maintenance Requirement (CMR): A CMR is a required task, established during the design certification of the airplane as an operating limitation of the type certificate. CMR's usually result from an analysis conducted to show compliance with catastrophic and hazardous failure conditions. Their intent is to detect significant latent failures that would, in combination with one or more other specific failures or events, result in a hazardous or catastrophic failure condition.

Structural Maintenance (ALI): The identification of Fatigue inspection tasks for each Principal Structural Element (PSE) is an integral part of certification for a damage tolerant aircraft. Transport Canada (TC) has a requirement that damage tolerant PSE's cannot go uninspected for more than half of the aircraft design lifetime (40,000 flights).

Safe Life Component (Discard Tasks): Operators must monitor the accumulated flight cycles and generate task cards to replace the safe life component, listed in the SAFE LIFE COMPONENTS or SUPPLEMENTS section, at or before a Safe Life Component reaches its Discard Time.

Fuel System Limitations (FSL): This section gives a list of the Fuel System Limitations (FSL) that are mandated by the Special Federal Aviation Regulation (SFAR) 88. FSLs are limitation requirements that identify all the necessary maintenance and inspection tasks to maintain the design features required to preclude the existence or development of an ignition source within the fuel tank system.

Critical Design Configuration Control Limitations (CDCCL): Design features that must be preserved to prevent the occurrence of an unsafe condition that was identified in the fuel tank assessment analysis (SFAR 88). The purpose of the CDCCL is to provide instructions to retain the critical ignition source prevention feature during configuration change that may be caused by alterations, repairs or maintenance actions.